



# CPSC 100

## Computational Thinking

### Artificial Intelligence (cont'd)

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# Agenda

- Course Admin
- Start Data Mining section!
- Introduction to Clustering
  - KMeans
  - DBScan

# Course Admin



# Admin

- **Reminder:** Milestone 4 is due today at 6 PM (48hr grace period exists if you need it)
- **Reminder:** Lab 8 is due Wednesday



# Admin

## Another Test!?

|                          |                                                             |                                         |                                            |                           |                           |
|--------------------------|-------------------------------------------------------------|-----------------------------------------|--------------------------------------------|---------------------------|---------------------------|
| <input type="checkbox"/> | <a href="#">Test 10</a> <input checked="" type="checkbox"/> | Yes <input checked="" type="checkbox"/> | 50 min <input checked="" type="checkbox"/> | 2026-03-28 10:00:00 (PDT) | 2026-04-01 23:59:00 (PDT) |
| <input type="checkbox"/> | <a href="#">Test 12</a> <input checked="" type="checkbox"/> | Yes <input checked="" type="checkbox"/> | 50 min <input checked="" type="checkbox"/> | 2026-03-28 10:00:00 (PDT) | 2026-04-10 23:59:00 (PDT) |
| <input type="checkbox"/> | <a href="#">Test 11</a> <input checked="" type="checkbox"/> | Yes <input checked="" type="checkbox"/> | 50 min <input checked="" type="checkbox"/> | 2026-04-06 10:00:00 (PDT) | 2026-04-10 23:59:00 (PDT) |

| Item               | Weight |
|--------------------|--------|
| Learning Logs (10) | 7%     |
| Labs (9)           | 18%    |
| Term Project (5)   | 25%    |
| Tests (10)         | 50%    |



# Learning Goals

After this week's lecture, you should be able to:

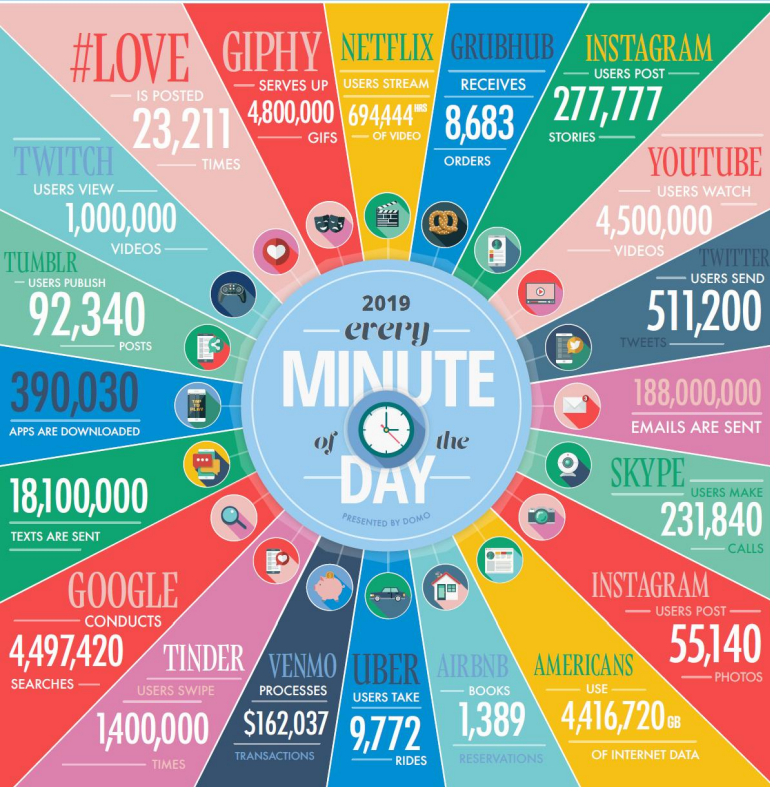
- Explain what is data mining, and why it is important to be aware of
- Differentiate between the uses of clustering and classification techniques and give examples of each.
- Define and identify the measures of quality for clustering algorithms.
- Convert between different data units (KB, MB, TB, etc...)



# DATA NEVER SLEEPS 7.0

How much data is generated *every minute*?

There's no way around it: big data just keeps getting bigger. The numbers are staggering, and they're not slowing down. By 2020, there will be 40x more bytes of data than there are stars in the observable universe. In our 7th edition of Data Never Sleeps, we bring you the latest stats on how much data is being created in every digital minute.



SOURCES: STATISTA, INTERNET LIVE STATS, EXPANDED RAMBLINGS, NATIONAL ASSOCIATION OF CITY TRANSPORTATION OFFICIALS, WIREX

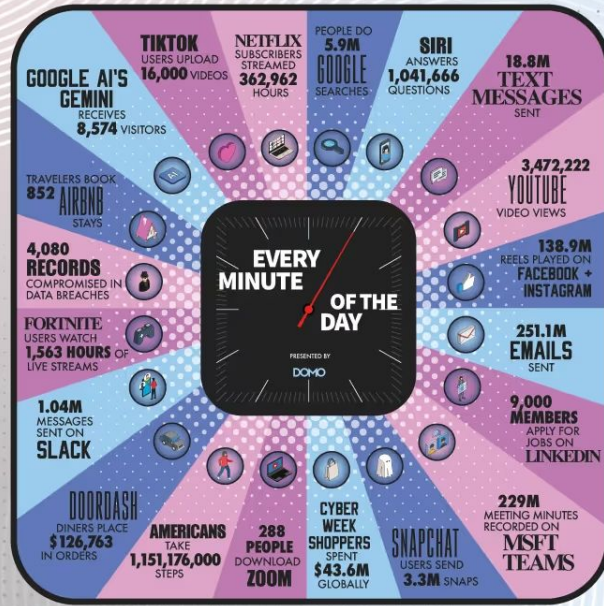


# DATA NEVER SLEEPS 12



Every minute of every day, the world generates a dizzying amount of data, and how we interact with it is constantly changing. AI tools are now answering millions of questions in real time and are transforming how we work, shop, and connect. Digital platforms are seeing explosive usage, with billions of emails, texts, and reels shared every day. Entertainment continues to drive engagement across streaming, gaming, and social media while e-commerce is setting new benchmarks as digital habits evolve and expand at an unprecedented pace.

In Domo's 12th edition of Data Never Sleeps, we capture a snapshot of this world powered by the rapid rise of data, AI, and digital activity, shaping every moment of modern life.



The world's internet population continues to grow significantly year-over-year. As of late 2024, 5.52 billion people—approximately 67.5% of the global population—are online.

According to industry analysts, the total amount of data created, captured, copied, and consumed globally is expected to reach 149 zettabytes by the end of 2024, with projections surpassing 314 zettabytes by 2028.

**Global Internet Population Growth**  
(IN BILLIONS)



As the volume and complexity of data accelerates, business success increasingly depends on the ability to turn information into insights. Domo helps you harness the power of data and AI so you can adapt as quickly as the world changes and make data-driven decisions that set you apart. Let Domo help you make sense of all the clicks, swipes, and streams so you can see the big picture shaped by every small decision.

Learn more at [domo.com](https://domo.com)

SOURCES: EARTHPIXL, DOTH STUDY, DEMANDBASE, HOOTSUITE, BUSINESSWIRE, DOORDASH, SQUARESPACE, 10TH ANNUAL DIGITAL INNOVATION, THUNDERSCAN, STATISTA, PIR NEWS, WIREX, TIKTOK

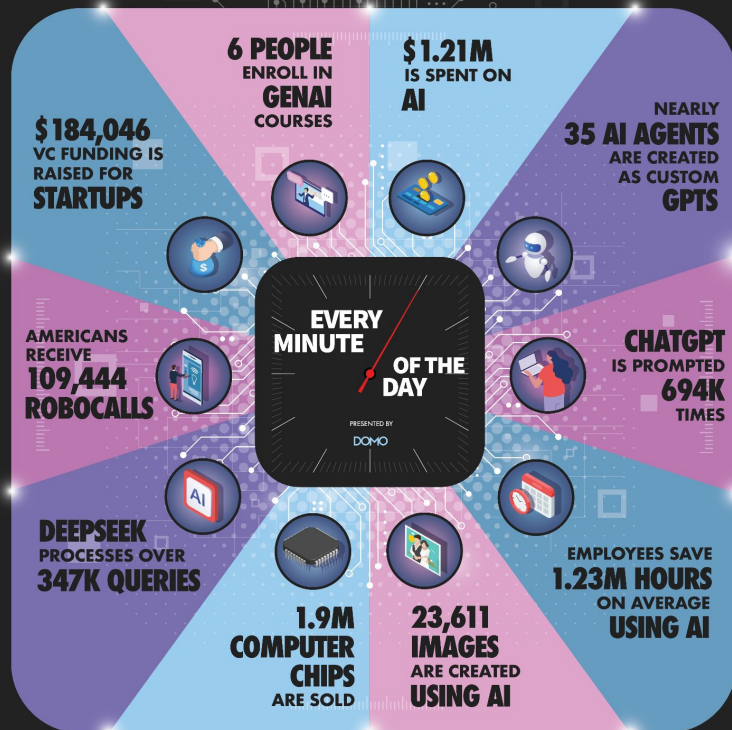


Source

# DATA NEVER SLEEPS AI EDITION 2025



Domo has tracked minute-by-minute digital engagement for over a decade, and AI is moving from experimental to essential at unprecedented speed. Spending on artificial intelligence has more than tripled since 2024. From boardrooms to living rooms, this special second Data Never Sleeps AI edition reveals how AI is reshaping the global economy, 60 seconds at a time.



Sources: Coolest Gadgets, Coursera, Crookin, Entrepreneur, Everquest, Gold IRA Companies, OpenAI, Precedence Research, SIA, Trading Economics

## DEMYSTIFYING DATA UNITS

From the more familiar 'bit' or 'megabyte', larger units of measurement are more frequently being used to explain the masses of data

| Unit                | Value           | Size                                    |
|---------------------|-----------------|-----------------------------------------|
| <b>b</b> bit        | 0 or 1          | 1/8 of a byte                           |
| <b>B</b> byte       | 8 bits          | 1 byte                                  |
| <b>KB</b> kilobyte  | 1,000 bytes     | 1,000 bytes                             |
| <b>MB</b> megabyte  | $1,000^2$ bytes | 1,000,000 bytes                         |
| <b>GB</b> gigabyte  | $1,000^3$ bytes | 1,000,000,000 bytes                     |
| <b>TB</b> terabyte  | $1,000^4$ bytes | 1,000,000,000,000 bytes                 |
| <b>PB</b> petabyte  | $1,000^5$ bytes | 1,000,000,000,000,000 bytes             |
| <b>EB</b> exabyte   | $1,000^6$ bytes | 1,000,000,000,000,000,000 bytes         |
| <b>ZB</b> zettabyte | $1,000^7$ bytes | 1,000,000,000,000,000,000,000 bytes     |
| <b>YB</b> yottabyte | $1,000^8$ bytes | 1,000,000,000,000,000,000,000,000 bytes |

\*A lowercase "b" is used as an abbreviation for bits, while an uppercase "B" represents bytes.

# A DAY IN DATA

The exponential growth of data is undisputed, but the numbers behind this explosion – fuelled by internet of things and the use of connected devices – are hard to comprehend, particularly when looked at in the context of one day

**500m**

tweets are sent every day

Twitter

**294bn**

billion emails are sent

Radicati Group

**3.9bn**

people use emails

**320bn**

emails to be sent each day by 2021

**306bn**

emails to be sent each day by 2020



**4PB**

of data created by Facebook, including

**350m** photos

**100m** hours of video watch time

Facebook Research



**4TB**

of data produced by a connected car

Intel

ACCUMULATED DIGITAL UNIVERSE OF DATA

**4.4ZB**

2013

**44ZB**

2020

PwC

From the more familiar 'bit' or 'megabyte', larger units of measurement are more frequently being used to explain the masses of data

| Unit                | Value                    | Size                                    |
|---------------------|--------------------------|-----------------------------------------|
| <b>b</b> bit        | 0 or 1                   | 1/8 of a byte                           |
| <b>B</b> byte       | 8 bits                   | 1 byte                                  |
| <b>KB</b> kilobyte  | 1,000 bytes              | 1,000 bytes                             |
| <b>MB</b> megabyte  | 1,000 <sup>3</sup> bytes | 1,000,000 bytes                         |
| <b>GB</b> gigabyte  | 1,000 <sup>3</sup> bytes | 1,000,000,000 bytes                     |
| <b>TB</b> terabyte  | 1,000 <sup>3</sup> bytes | 1,000,000,000,000 bytes                 |
| <b>PB</b> petabyte  | 1,000 <sup>3</sup> bytes | 1,000,000,000,000,000 bytes             |
| <b>EB</b> exabyte   | 1,000 <sup>3</sup> bytes | 1,000,000,000,000,000,000 bytes         |
| <b>ZB</b> zettabyte | 1,000 <sup>3</sup> bytes | 1,000,000,000,000,000,000,000 bytes     |
| <b>YB</b> yottabyte | 1,000 <sup>3</sup> bytes | 1,000,000,000,000,000,000,000,000 bytes |

\*A lowercase "b" is used as an abbreviation for bits, while an uppercase "B" represents bytes.

**65bn**

messages sent over WhatsApp and two billion minutes of voice and video calls made

Facebook



Searches made a day

**5bn**

Searches made a day from Google

**3.5bn**

Smart Insights



**463EB**

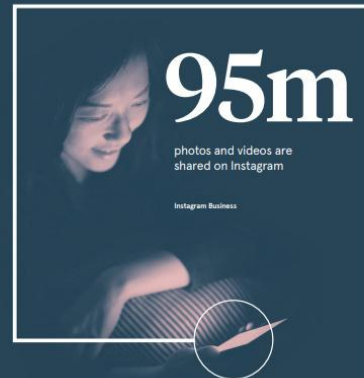
of data will be created every day by 2025

IDC

**95m**

photos and videos are shared on Instagram

Instagram Business



**28PB**

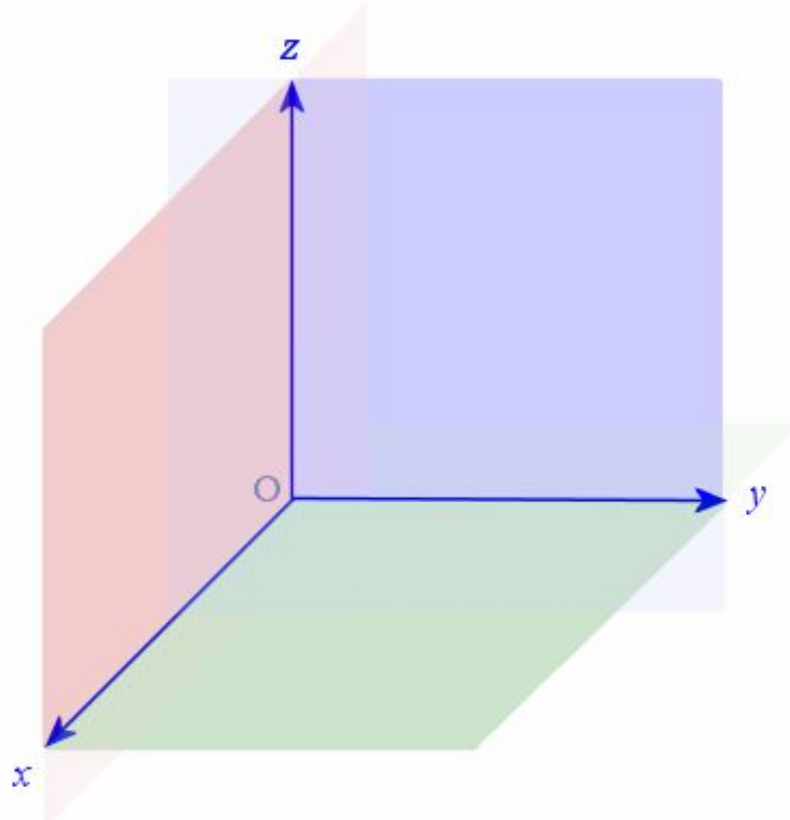
to be generated from wearable devices by 2020

Statista



Source **RACONTEUR**

# Data Mining - Personal Data



# Why Should I Care about Data Mining?

## Humanity Passes 1 Zettabyte Mark in 2010

A zettabyte is 1,000,000,000,000,000,000 bytes (that's 21 zeroes for those counting), or one trillion gigabytes. That's enough data to fill 75 billion 16-gigabyte-sized iPads.



By 2020, about 1.7MB of new data will be created every second for every human on the planet.

By then, our accumulated digital universe of data will grow from 4.4 zettabytes today to ~44 zettabytes, or 44 *trillion* gigabytes.

A trillion gigabytes is a **zettabyte** ,  
1 000 000 000 000 000 000 000 bytes  
 $10^{21}$  bytes  
 $2^{70}$  bytes



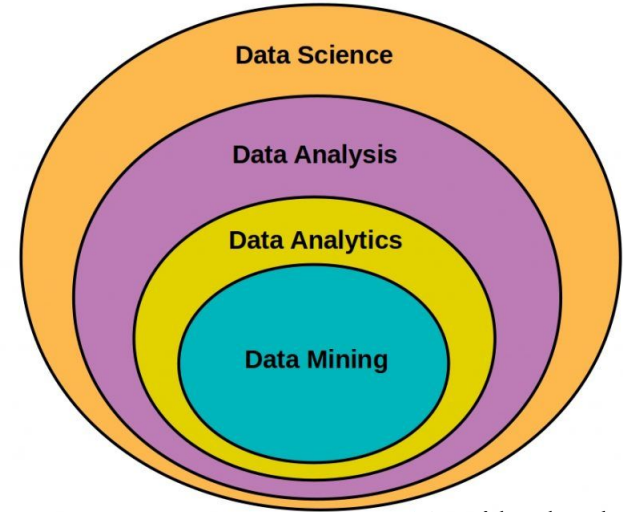
# Data Mining –Definition

Data mining is the process of looking for patterns in large data sets

There are many different kinds for many different purposes

In depth exploration of some tasks:

- Clustering (k-means and DBScan)
- Classification (decision trees)
- Also look at kNN



# Clustering Definition



Clustering is partitioning a set of items into subgroups so as to ensure certain measures of quality (e.g., “similar” items are grouped together)

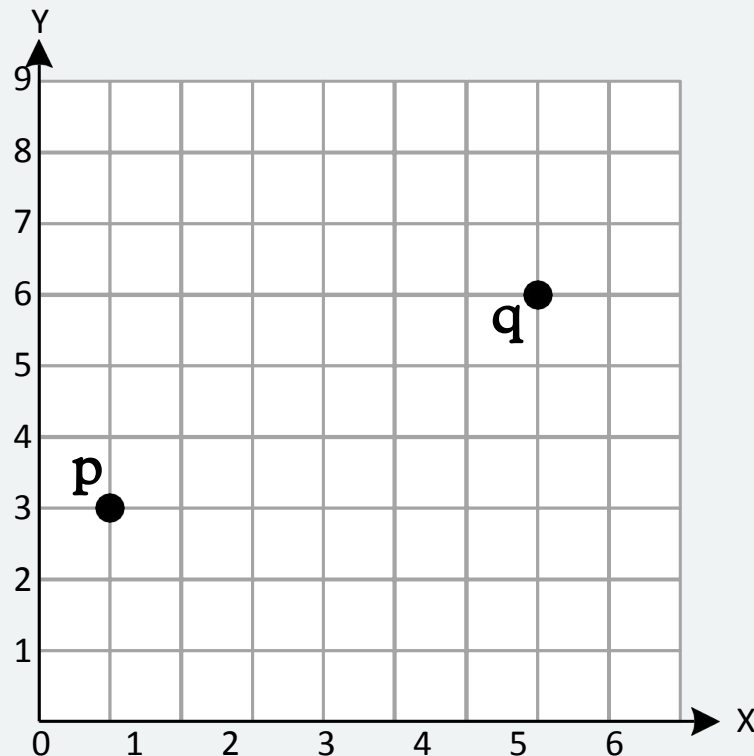
“Similar”, for our purposes is measured by the Euclidean distance between the points

$$d(p, q) = \sqrt{\sum_{i=1}^n (q_i - p_i)^2}$$

$p, q$  = Two points in Euclidean n-space

$q_i - p_i$  = Euclidean vectors

$n$  = n-space



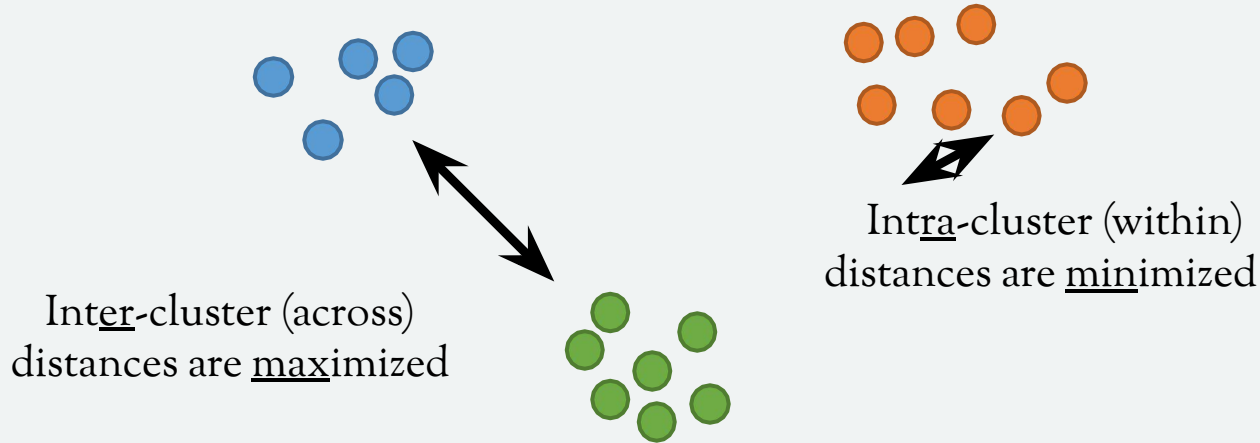
# KMeans Algorithm - Watch Video before next class!

K-means Clustering!!!

# What is Good Clustering?

Items in the *same* cluster are similar (*intra-cluster*)

Items in *different* clusters are different  
(*inter-cluster*)



# Measuring Clustering Quality Possible Criteria to Use

**Intra-class (intra-cluster) similarity:** points within a cluster contain are close to each other (or at least to their closest neighbours)

**Inter-class (inter-cluster) dissimilarity:** points in two different clusters are far from each other (or at least to their closest neighbours in other clusters)

**Size similarity :** clusters have similar size

# Wrap up